

More than Fitness:

User Perceptions and Hopes for Getting Fit in Virtual Reality

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ABSTRACT

Fitness activities are linked to health, cognitive acuity, and the brain's ability to respond to a stimulus. This article reports on a mixed methods investigation of user experiences of a commercial off the shelf virtual reality fitness experience. A total of 74 visitors to a southern science museum, spanning from under 18 years of age to 64 years of age participated in a VR fitness experience and completed a questionnaire and a semi-structured interview. Participants reported experiencing social presence with pre-recorded fitness instructors, positive usability, and a majority predicted that using the tool could improve cognitive acuity. This pilot indicates a cross generational interest in VR fitness tools.

Keywords: Virtual Reality, Physical activity intervention, User Experience Testing, UX, Diverse Population

1 INTRODUCTION

While people frequently think of VR headsets as gaming devices, many consumers use VR for many experiences beyond gaming, such as attending concerts, thrill seeking, trying on clothing, shoes, or makeup, or fitness tools [2]. As of the writing of this manuscript, the Meta Quest store, where people discover software experiences for the Meta Quest, Meta Quest Pro, and Oculus Rift, has over 30 fitness apps available for consumer purchase. Many of these apps provide experiences that allow users to do activities from meditation to dancing, to boxing, while immersed real and imagined remote locations.

VR experiences have been shown to encourage and increase physical activity among young and older people and advance not only physical health of users, but also their mental well-being [3]. Studies have shown that viewing nature in virtual reality has a positive effect on those confined to assisted living settings. Similarly, studies have shown that viewing nature in virtual reality has a positive effect on those confined to assisted living settings [4]. While existing research focuses primarily on the implications of VR in labs and clinical settings for rehabilitation and research, the propagation of VR technologies in consumer's homes provides an opportunity to explore the impact of off the shelf virtual reality fitness tools.

A scoping review of forty articles [5], found positive correlations between physical activity and cognitive skills. Yet understanding the benefits of physical activity does not always motivate individuals to exercise. Active video games that are presented through virtual reality (VR) have been seen to build motivation while increasing the intensity of physical activity [6].

This pilot study is the first stage in a long term study into the effectiveness for commercial VR into improving cognitive function in older adults. To that end, our research questions are:

RQ1: Do users perceive the VR fitness as useful for cognitive function?

RQ2: Do users feel a sense of social presence with the instructor?

RQ3: Do users feel like they would use the tool often?

2 TESTING COMMERCIAL OFF THE SHELF VR EXPERIENCES

The pilot study reported in this manuscript introduced people to a novel VR fitness experience and conducted a mixed methods exploration of individual attitudes about the satisfaction, perceived usefulness, and interest in using in the future. This pilot study was conducted in a science museum in North Texas. Participants were museum visitors who were invited to try a fitness app in VR. The participants were given informed consent forms and minors also had to have parents complete an assent form. After completing consent forms, participants were introduced to the Meta Quest and then placed in a short experience designed to introduce them to the experience with very little instruction from the research team.

2.1 Supernatural Fitness

We chose the virtual fitness experience, Supernatural Fitness for our research. Supernatural is a highly immersive interactive experience [7] that immerses users in novel 360 degree environments in a head mounted display, while engaging the user's upper and lower body in squats, lunges, and arm movements, tracked by VR controllers.



Figure 1: Supernatural Fitness Interactions

2.2 Participants

A total of 74 museum-goers agreed to participate in the study. Participants ranged in age from under 18 years to age 64; with 39 under 18; eight participants between 18 and 24; nine participants between 24 and 34; 11 between the ages of 35 – 44; two between the ages of 45 and 54, and three participants above the age of 64. The study's participant composition included 38 individuals self-identified as White / Caucasian, 17 participants of Hispanic origin, 6 individuals of Asian - Indian descent, 3 Native-American participants, 8 individuals identifying as Black or African American, 2 participants of Asian - Eastern heritage, 2 individuals of mixed race, and 1 participant identifying as American Indian or Alaska Native; 7 participants chose not to disclose their ethnicity. Only 8 of the 74 participants owned a headset.

2.3 Study Design

After completing informed consent, participants were placed in the Oculus Quest set up with a six-minute tutorial introducing them to supernatural fitness environment and the workout

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activities. After completing the tutorial, the VR headset was removed and the participant responded to a 20-item questionnaire and engaged in a short, semi-structured interview.

3 RESULTS

Of the 74 participants, 49 completed the semi-structured interviews and 58 completed the questionnaire, including the System Usability Scale (SUS) [8]. Responses for the semi-structured interviews were written and transcribed into an excel sheet for qualitative analysis in the qualitative software tool, nVivo.

For RQ1: Do users perceive the VR fitness as useful for cognitive function, we looked at the responses to the semi-structured interview question, “Do you feel like this would affect your cognitive acuity (mental sharpness) in any way if you used it more regularly?” 21 participants (42%) indicated, “Yes,” it would impact cognitive acuity, nine participants (18%) said it would not, and 13, (40%) were unsure.

We investigated RQ2: Do users feel a sense of social presence with the instructor? With two semi structured interview questions asking what the user thought about the instructor. Answers that indicated the user felt the instructor was real and or they felt connected with them, that user was coded as socially present with the instructor. 72% of participants indicated affinity and connection with the instructor. 42% of users indicated they would engage with the instructor in real life, if given the opportunity. One participant noted, for example, “Yes, I love working out and I love helping my body. I would love to get tips from her about being healthy and how much water to drink.”

Finally, we used the question, “How often do you think you’d use this tool if you had access to it?” to analyze RQ3: Do users feel like they would use the tool often? 18 participants (37%) indicated they would use the tool daily, while 28 participants (57%) predicted they would use it between once and five times a week. Interestingly, only three participants indicated they would never use the tool, even if they had access to it.

4 DISCUSSION

The current study explored user perceptions of VR fitness in three key areas: cognitive function, social presence with the instructor, and anticipated tool usage. While 72% felt an affinity with the virtual instructor with some indicating they would want to engage with them in “real life” (42%). This is particularly interesting, given that the fitness instructors are pre-recorded and don’t ever actively engage with the users. Future research might also leverage the Supernatural Fitness Facebook community with which users can engage. As of the writing of this manuscript, there were over 92 thousand members in the “Official Supernatural Community” Facebook page. People connect with existing friends, share their workouts, ask questions, and even start group challenges with people in the group. These challenges further engage the gamification qualities of the app.

While 40% of users reported that they would use the tool daily, it will be interesting to discover actual engagement users have in a long-term study. and only 3% showed no interest in using it. These results suggest a favorable inclination towards VR fitness, indicating its potential for enhancing cognitive function and fostering engagement through strong social connections with virtual instructors. Finally, the analysis of the open-ended responses revealed that 42% of participants believed regular use of VR fitness would positively impact cognitive acuity. One participant noted, “I think it would affect it in a good way affecting eye/hand coordination.”

5 CONCLUSION

This pilot reveals that people are getting more than just a fitness experience from virtual reality fitness tools. Next steps for our research include a longitudinal study on the adoption of a virtual reality fitness experience across age ranges. We will work to recruit a diverse participant pool across age ranges and physical and mental ability. In this long term study we will further investigate the concepts discussed in this paper. 1) Do these experiences impact cognitive ability? 2) Will people use the apps as often as they predict? 3) What impact, if any, will the sense of connection with an asynchronous recording have on a user’s sense of well-being and motivation to continue engaging with the Virtual Reality Fitness Experience. Future research should leverage the capacity of these technologies to offer real-time data sharing and big data analysis, and enhancing participation through tools like heart rate recording, social media integration, and social interaction [9]. To this end, our future work will integrate the capacity of supernatural fitness to report performance in terms of accuracy, speed, and physiological data such as heart rate. The combination of this data has the potential to provide robust insights into the impact of these emerging experiences for diverse groups of users, across user types, demographics, and use cases.

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REFERENCES

- [1] Richter, F. (October 14, 2022). AR & VR Adoption Is Still in Its Infancy.
- [2] Sitecore. (October 19, 2022). Digital activities that metaverse enthusiasts in the United States want to use the metaverse for as of August 2022 [Graph]. In Statista. Retrieved January 21, 2023.
- [3] Deutsch, J., & McCoy, S. W. (2017). Virtual reality and serious games in neurorehabilitation of children and adults: Prevention, plasticity and participation. *Pediatric physical therapy : The official publication of the Section on Pediatrics of the American Physical Therapy Association*, 29, S23.
- [4] Yu, C.-P., Lee, H.-Y., Lu, W.-H., Huang, Y.-C., & Browning, M. H. E. M. (2020). Restorative effects of virtual natural settings on middle-aged and elderly adults. *Urban Forestry & Urban Greening*, 56, 126863. <https://doi.org/10.1016/j.ufug.2020.126863>
- [5] Erlenbach, E., McAuley, E., & Gothe, N. P. (2021). The association between light physical activity and cognition among adults: a scoping review. *The Journals of Gerontology: Series A*, 76(4), 716-724.
- [6] Gao, Z., & Lee, J. E. (2019). Emerging technology in promoting physical activity and health: Challenges and opportunities. *Journal of Clinical Medicine*, 8(11), 1830. <https://doi.org/10.3390/jcm8111830>
- [7] Hayes, A., Dhimolea, Meng, N., and Tesh, G. (2021). Levels of immersion for language learning from 2D to highly immersive interactive VR.
- [8] Brooke, J. (1996). SUS-A quick and dirty usability scale. *Usability evaluation in industry*, 189(194), 4-7.
- [9] Dębska, M., Polechoński, J., Mynarski, A., & Polechoński, P. (2019). Enjoyment and intensity of physical activity in immersive virtual reality performed on innovative training devices in compliance with recommendations for health. *International journal of environmental research and public health*, 16(19), 3673.